Chapter 2. Outliers and Robust Prediction

Predictive Modeling, Section 2 (MCD, LOF, Isolation Forest, leverage, Cook's distance, Huber, RANSAC, Theil–Sen)

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1 An Outlier Is Not the Same as a Data Error

What the observation is	Example	First response
input / measurement error	wrong unit, mis-recorded target	correct from the raw source; remove only if uncorrectable
a genuine rare case	a very large transaction, a rare high-risk patient	keep it, and check that group's prediction error separately
a case from a new environment	a policy or market condition absent in training	check distribution shift and the need to retrain

An outlier is a *statement relative to a reference*, and the right action depends on why it is unusual.

1.1 Three kinds of outliers

Kind	Meaning	Simple diagnostic
univariate	one variable is far from most observations	histogram, boxplot, IQR rule
multivariate	each variable is ordinary but the *combination* is rare	scatterplot, multivariate distance, density
conditional	given x_i, the target y_i differs greatly from what is expected	residual $e_i = y_i - \hat{y}_i$

The IQR rule flags $x < Q_1 - 1.5\,\mathrm{IQR}$ or $x > Q_3 + 1.5\,\mathrm{IQR}$, $\mathrm{IQR} = Q_3 - Q_1$. Univariate rules alone miss outliers that appear only in combinations of variables.

Method	What it checks	Observations it looks for
MCD	global distance in X	multivariate outliers far from the central distribution
LOF	local density in X	points less dense than their neighbours
Isolation Forest	isolation path in X	points isolated by few random splits
studentised residual	y given X	cases far from the regression prediction
leverage, Cook's distance	influence on the regression fit	cases that change coefficients and predictions a lot

"X is unusual" and "y is unusual given X" are different problems.

2 MCD: Robust Mahalanobis Distance

2.1 Classical Mahalanobis distance and masking

With the sample mean $\bar{x}$ and sample covariance S,

$$D_i^2 = (x_i - \bar{x})^\top S^{-1} (x_i - \bar{x}).$$

Theorem: null distribution of the squared distance

If $X \sim N_p(\mu, \Sigma)$ then $(X-\mu)^\top \Sigma^{-1}(X-\mu) \sim \chi^2_p$. Hence, approximately, a point with $D_i^2 > \chi^2_{p,1-\alpha}$ is an outlier candidate.

Proof. $Z = \Sigma^{-1/2}(X-\mu) \sim N_p(0, I_p)$, and $(X-\mu)^\top\Sigma^{-1}(X-\mu) = Z^\top Z = \sum_{j=1}^p Z_j^2$, a sum of p independent squared standard normals. ■

Masking. Outliers pull $\bar{x}$ toward themselves and inflate S, so their own distances shrink and they may no longer be flagged. The remedy is to estimate location and scatter from the bulk of the data only.

2.2 Minimum Covariance Determinant

Definition: MCD

For a fixed h with $n/2 \le h < n$ (typically $h \approx 0.75n$), find the subset H of h observations whose sample covariance has the smallest determinant:

$$H^\star = \arg\min_{|H|=h} \det S_H, \qquad \hat{\mu}_{\mathrm{MCD}} = \bar{x}_{H^\star}, \quad \widehat{\Sigma}_{\mathrm{MCD}} = c_{h,n}\, S_{H^\star},$$

($c_{h,n}$ is a consistency factor), and use the robust distance

$$D_i^2 = (x_i - \hat{\mu}_{\mathrm{MCD}})^\top \widehat{\Sigma}_{\mathrm{MCD}}^{-1} (x_i - \hat{\mu}_{\mathrm{MCD}}), \qquad \text{flag if } D_i^2 > \chi^2_{p,1-\alpha}.$$

$\det S_H$ is proportional to the squared volume of the data ellipsoid of H, so MCD looks for the *tightest* h points; outliers enlarge the volume and are excluded. Up to $n-h$ outliers cannot drive the estimate arbitrarily far (breakdown point $\approx (n-h)/n$). In practice the FAST-MCD algorithm iterates "C-steps": given H, compute distances and replace H by the h points with smallest distances; each step does not increase $\det S_H$.

Remark: when MCD fits

MCD suits continuous data with $p < n$ and a single elliptical bulk. With multimodal or nonlinear clouds, local methods (LOF) or Isolation Forest are more appropriate.

3 LOF: Lower Density Than the Neighbours

Definition: local outlier factor

Let $N_k(i)$ be the k nearest neighbours of i, $d_k(j)$ the distance from j to its kth neighbour, and the reachability distance $\mathrm{rd}_k(i,j) = \max\{d_k(j), d(i,j)\}$. The local reachability density and LOF are

$$\mathrm{lrd}_k(i) = \Big(\frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \mathrm{rd}_k(i,j)\Big)^{-1}, \qquad \mathrm{LOF}_k(i) = \frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \frac{\mathrm{lrd}_k(j)}{\mathrm{lrd}_k(i)}.$$

$\mathrm{lrd}_k(i)$ grows as the distances to the neighbours shrink. LOF ≈ 1: density similar to the neighbours; LOF > 1: lower density than the neighbours; very large LOF: a local outlier separated from its surrounding group. Because the comparison is *relative to the neighbourhood*, a point in a sparse but genuine cluster is not penalised, unlike a global distance rule. The reachability distance smooths $d(i,j)$ for very close pairs, which stabilises the ratio.

4 Isolation Forest: Quickly Isolated Points

Algorithm: Isolation Forest

1. On a subsample, choose a feature uniformly at random.
2. Choose a split point uniformly between the feature's minimum and maximum.
3. Split recursively until each observation is alone (or a depth limit).
4. Grow many such trees and average the path lengths.

Anomaly score:

$$s(x) = 2^{-\mathbb{E}[h(x)]/c(n)}, \qquad c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \quad H(m) \approx \ln m + 0.5772,$$

where $h(x)$ is the path length until x is isolated and $c(n)$ is the average path length of an unsuccessful search in a binary search tree of n points (normalisation by sample size).

Short paths give large scores: $s \to 1$ means anomalous, $s \approx 0.5$ for all points means no clear anomaly, $s \ll 0.5$ means normal. No distributional shape is assumed, and it scales to large data. The parameter `contamination` only sets the share of points to be labelled anomalous. Isolation Forest gives a score, but it does not tell a data error from a valid rare case.

5 Regression: Residual, Leverage and Influence

5.1 Three different notions

Suppose most data follow $y \approx 2x$.

Observation	Feature x	Residual	Interpretation
$(3, 20)$	ordinary range	large	outlier in the target
$(10, 20)$	far from the rest	small	high-leverage point that follows the trend
$(10, 0)$	far from the rest	large	influential point likely to change the line

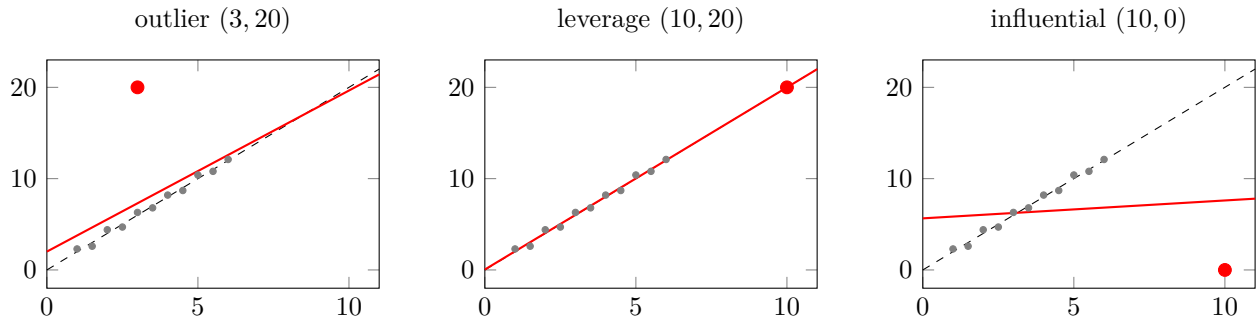


Figure 1: Dashed: OLS line on the grey points ($y \approx 2x$); red: OLS line after adding the red point (exact refits). A target outlier in the middle of the x-range mostly shifts the line upward; a high-leverage point on the trend changes nothing; a high-leverage point off the trend drags the slope — large residual *and* large leverage, hence large Cook's distance.

The residual is the gap between truth and prediction; leverage measures how unusual x_i is in feature space; an influential point is one whose removal changes the fit a lot.

5.2 Hat matrix and leverage

With $X \in \mathbb{R}^{n\times(p+1)}$ (intercept included) and $H = X(X^\top X)^{-1}X^\top$, $\hat{y} = Hy$ and $e = (I-H)y$. The leverage is $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$.

Theorem: properties of leverage and residuals

1. $0 \le h_{ii} \le 1$ and $\sum_i h_{ii} = p+1$, so the average leverage is $(p+1)/n$.
2. $\hat{y}_i = \sum_j h_{ij} y_j$, so $h_{ii} = \partial \hat{y}_i / \partial y_i$.
3. Under $\mathrm{Var}(\varepsilon) = \sigma^2 I$, $\mathrm{Var}(e_i) = \sigma^2(1-h_{ii})$.

Proof. H is symmetric idempotent, so $h_{ii} = (H^2)_{ii} = \sum_j h_{ij}^2 \ge h_{ii}^2$, giving $0 \le h_{ii} \le 1$; $\sum_i h_{ii} = \mathrm{tr}H = \mathrm{tr}\{(X^\top X)^{-1}X^\top X\} = p+1$. Item 2 is $\hat{y} = Hy$. For 3, $e = (I-H)\varepsilon$ because $(I-H)X = 0$, so $\mathrm{Var}(e) = \sigma^2(I-H)(I-H)^\top = \sigma^2(I-H)$. ■

A high-leverage point pulls the fitted line toward itself, so its *raw* residual tends to be small. Item 3 is why residuals must be standardised.

5.3 Studentised residual and Cook's distance

Diagnostic	Computation	Cases it reveals
studentised residual	$r_i = \dfrac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}$	y_i far from what x_i predicts
leverage	h_{ii}, diagonal of H	far from the other observations in feature space
Cook's distance	$D_i = \dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$	removing it changes coefficients and predictions much

Common screening rules are $|r_i| > 2$, $h_{ii} > 2(p+1)/n$, $D_i > 4/n$. Exceeding a rule makes a case *an object of inspection*, not an automatic deletion.

Theorem: leave-one-out update and Cook's formula

Let $\hat{\beta}_{(i)}$ be the OLS estimate without case i. Then

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{(X^\top X)^{-1} x_i e_i}{1-h_{ii}},$$

and Cook's distance, defined as $D_i = \dfrac{\|X\hat{\beta} - X\hat{\beta}_{(i)}\|^2}{(p+1)\hat{\sigma}^2}$, equals $\dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$.

Proof. Let $A = X^\top X$. Removing row i gives $A_{(i)} = A - x_i x_i^\top$ and $X_{(i)}^\top y_{(i)} = X^\top y - x_i y_i$. Sherman–Morrison: $A_{(i)}^{-1} = A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}$. Then

$$\hat{\beta}_{(i)} = \Big(A^{-1} + \tfrac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}\Big)(X^\top y - x_i y_i) = \hat{\beta} - A^{-1}x_i y_i + \frac{A^{-1}x_i(\hat{y}_i - h_{ii}y_i)}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}x_i(y_i - \hat{y}_i)}{1-h_{ii}}.$$

Hence $\|X(\hat{\beta} - \hat{\beta}_{(i)})\|^2 = \frac{e_i^2}{(1-h_{ii})^2} x_i^\top A^{-1} X^\top X A^{-1} x_i = \frac{e_i^2 h_{ii}}{(1-h_{ii})^2}$. Divide by $(p+1)\hat{\sigma}^2$ and substitute $e_i^2 = r_i^2\hat{\sigma}^2(1-h_{ii})$. ■

So Cook's distance is large only when *both* the standardised residual and the leverage factor $h_{ii}/(1-h_{ii})$ are sizeable, which is exactly the case $(10, 0)$ above.

Remark: leave-one-out residuals for free

The same identity gives $y_i - x_i^\top \hat{\beta}_{(i)} = e_i/(1-h_{ii})$, so the leave-one-out CV error of OLS is $\frac{1}{n}\sum_i \{e_i/(1-h_{ii})\}^2$ without refitting.

6 What to Do With Detected Cases

Finding	Action	Caveat
input / measurement error	correct or remove in raw data	remove only if it cannot be corrected and cannot be used
heavily skewed variable	log or Box–Cox transform	keeps the order, compresses gaps among large values
extremes must be capped	winsorisation	cap both tails at quantiles *estimated on training data*
valid rare case	keep	use robust learning and segment-wise evaluation

Never delete or alter observations on the basis of a detection score alone.

7 Robust Learning

Method	Core principle	Suitable when
Huber regression	caps the growth rate of loss for large residuals	a few large target outliers
RANSAC	fits many small subsamples; keeps the model supported by most points	clear inlier vs. contaminated groups
Theil–Sen	median of pairwise slopes	linear regression with few features

7.1 Huber loss

With residual $r = y - \hat{y}$:

Loss	Definition	Large residual	Target of prediction
squared	r^2	grows quadratically	conditional mean
absolute	$\lvert r\rvert$	grows linearly	conditional median
Huber	quadratic for small, linear for large $\lvert r\rvert$	linear beyond δ	compromise between centre fit and tail robustness

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta(|r| - \frac{1}{2}\delta), & |r| > \delta, \end{cases} \qquad \psi_\delta(r) = \rho'_\delta(r) = \begin{cases} r, & |r| \le \delta, \\ \delta\,\mathrm{sign}(r), & |r| > \delta. \end{cases}$$

Derivation: why Huber limits outliers — IRLS view

The estimating equation is $\sum_i \psi_\delta(y_i - x_i^\top\beta)x_i = 0$. Writing $\psi_\delta(r) = w(r)\,r$ with $w(r) = \min\{1, \delta/|r|\}$ turns it into weighted least squares $\sum_i w_i(y_i - x_i^\top\beta)x_i = 0$: iterate $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$ with $W = \mathrm{diag}\{w(r_i)\}$. A case with residual 10δ gets weight 0.1; its pull on the fit (the influence ψ) is bounded by δ, whereas under squared loss it grows without bound. The function ρ_δ is convex and continuously differentiable, so the minimiser

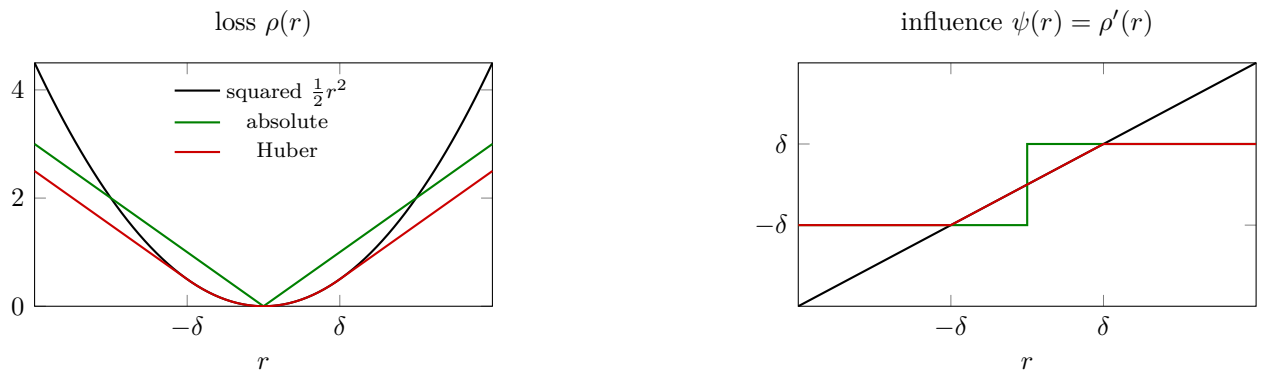


Figure 2: Squared, absolute and Huber losses (left) and their derivatives (right). The influence of a residual on the estimating equation grows without bound under squared loss but is capped at $\pm\delta$ under Huber loss.

is well defined. A smaller δ limits large residuals more strongly; as $\delta \to \infty$ Huber becomes OLS, as $\delta \to 0$ it behaves like LAD (rescaled).

Since δ is in units of y, it is usually set as $\delta = 1.345\hat{s}$ with a robust scale $\hat{s}$ (e.g. MAD/0.6745); 1.345 gives 95% efficiency under Gaussian errors.

7.2 RANSAC

Algorithm: RANSAC

1. Fit the regression on a small random subsample (minimal size s).
2. Classify observations with residual below a threshold as inliers.
3. Repeat many times.
4. Refit the model on the largest inlier set.

If a fraction w of points are inliers, a subsample of size s is all-inlier with probability w^s, so to succeed at least once with probability q one needs $N \geq \log(1-q)/\log(1-w^s)$ trials. The residual threshold and the number of trials affect the result.

7.3 Theil–Sen

In simple regression compute all pairwise slopes:

$$\hat{\beta}_1 = \underset{i<j}{\text{median}} \frac{y_j - y_i}{x_j - x_i}, \qquad \hat{\beta}_0 = \underset{i}{\text{median}}(y_i - \hat{\beta}_1 x_i).$$

A few extreme slopes cannot dominate the estimate; the breakdown point is about $1 - 1/\sqrt{2} \approx 29\%$. RANSAC separates inliers and outliers sharply; Theil–Sen robustifies continuously through the median of slopes.

7.4 Quantile regression

The pinball loss of Ch. 1 gives the conditional τ-quantile; $\tau = \frac{1}{2}$ is LAD (least absolute deviations). It is robust to outlying y (not to high-leverage x) and describes the conditional distribution beyond the centre.

8 Robust Prediction Is Also Chosen by CV

Algorithm: protocol

1. Split off the final test set first.
2. Inside each CV training fold, fit scaling and outlier-detection steps.
3. Set the detection threshold, LOF's k, Huber's δ, RANSAC's residual threshold from training folds only.
4. Compare OLS and robust regressions on the same validation folds.
5. Check errors of outlier candidates and of rare groups along with the overall error.

Result to check	Purpose
MAE and RMSE	evaluate typical errors and rare large errors together
sensitivity to the detection rule	does model selection change with the threshold?
error of rare groups	did robustification ignore important cases?

Summary

- Outliers are relative to a reference; errors, valid rare cases and new environments call for different actions.
- In X: MCD (global, elliptical), LOF (local density), Isolation Forest (isolation depth).
- In regression: residual (y given x), leverage (x unusual), Cook's distance (both together).
- Robust losses (Huber, LAD, quantile) and robust fitting (RANSAC, Theil–Sen) limit the dominance of a few points; their tuning constants are chosen inside CV.